

Literature Status for arXiv:2409.14481, Question 5.2

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Status

Status: literature already answered for Question 5.2 of arXiv:2409.14481. An adjacent subcase of the final similarity question is also answered: the case $q = 2$ is positive.

Source Question

The source paper is Valentin Gillet, *Typical properties of positive contractions and the invariant subspace problem*, arXiv:2409.14481. In Section 5, “Further remarks and questions”, Question 5.2 asks whether the point spectrum of an SOT-typical positive contraction on ℓ_2 is $\mathbb{D}$. The final question of that section asks whether SOT and SOT* are similar on $\mathcal{P}_1(\ell_q)$ for $1 < q \leq 2$.

Later Answer

The later paper is Valentin Gillet, *Similar operator topologies on the space of positive contractions*, arXiv:2412.11587. Its introduction states that it answers Question 5.2 from arXiv:2409.14481. The theorem labeled `introthvpspos`, restated in the applications section as `corvpspostopotau`, proves that for every Polish operator topology on $\mathcal{P}_1(\ell_2)$ lying between WOT and SOT*, a typical positive contraction T has no eigenvalue and T^* has no eigenvalue. In particular, for the SOT topology on $\mathcal{P}_1(\ell_2)$,

$$\sigma_p(T) = \emptyset$$

generically. Thus the answer to Question 5.2 is negative: the point spectrum is not $\mathbb{D}$.

The same paper’s theorem labeled `introthtoposimsur12` proves that all Polish operator topologies between WOT and SOT* are similar on $\mathcal{P}_1(\ell_2)$. In particular SOT and SOT* are similar on $\mathcal{P}_1(\ell_2)$, giving a positive answer to the $q = 2$ subcase of the source paper’s final similarity question.

Scope Limitations

This packet records a literature answer, not a new proof. It does not settle the invariant-subspace question for $1 < q \neq 2 < \infty$, the hypercyclicity of $(2T)^*$ for $1 < q < 2$, the non-Hilbert similarity cases, the SOT version of the reflexive-monotone-basis AAB theorem, or the compact-commutant question for typical positive contractions on ℓ_2 .

Search Evidence

The local run indexes had no prior row for arXiv:2409.14481. Exact web and local-source searches for the source title, Question 5.2, point spectrum of typical positive contractions, and similar operator topologies found arXiv:2412.11587 as the later answer. The supporting paper explicitly names Question 5.2 from the source paper, so the Question 5.2 part belongs in the literature-already-answered bucket.